

Predictions of DW-Nominate Scores of Given Tweets

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Abstract—This lab report is an executive summary of the creation of a model to predict the DW-Nominate scores, both 1st and 2nd dimension of political ideology, for unique tweets via Twitter. The final RMSE score of the model is 0.25667, which is eighth place in the Kaggle Competition as of 11:13 AM on February 9th.

I. INTRODUCTION

Using a dataset containing twitter data of unique tweets from 2008-2020, I used separate training and testing datasets to predict the 1st dimension of political ideology scores as well as the 2nd dimension of political identity scores for each testing dataset tweet based on certain categorical features.

The model created implements data preprocessing techniques to clean and lemmatize both the training and testing data tweets, making them more legible for the model. Also, the cleaned text was vectorized, modifying the column to become numerical data, becoming more model-friendly.

Through Autogluon's TabularPredictor model and Grid-SearchCV hyperparameter tuning techniques, a LightGBM model was utilized for this project.

The model takes approximately 30 minutes in total to preprocess, train, and predict DW-Nominate scores of the test dataset. Its best accuracy score on Kaggle is 0.25677.

The parameters for the LGM Model are:

- boosting_type: Gradient Boosting Decision Trees
- n_estimators: 4000
- learning_rate: 0.07
- n_jobs: -1

II. DATA

A. About the Dataset

The dataset, obtained via Kaggle, contains tweet data posted by United States Congress members through the years 2008-2020. Features included in the dataset for each tweet include: a unique ID number, the number of favorites, the full text of the tweet, the hashtags included, the amount of retweets, the year of the tweet, as well as the 1st and 2nd dimension of political ideology.

B. Training and Testing Split

The dataset is pre-sorted into two segments, training data and testing data. The training data consists of 333,987 tweets and the test data is 135,753 tweets, but omits the dimension 1 and dimension 2 nominate scores.

Metric	Minimum	Average	Median	Maximum
Tweet (Character Length)	7	183.176139	154.0	2437
Tweet (Word Count)	1	1.006072	1.0	7
Hashtag (Character Length)	1	14.536123	12.0	184
Hashtag (Word Count)	1	1.491980	1.0	17

Fig. 1. Statistics for Minimum, Maximum, Average, and Median Length of Training Data Tweets, for both tweet lengths and hashtag lengths

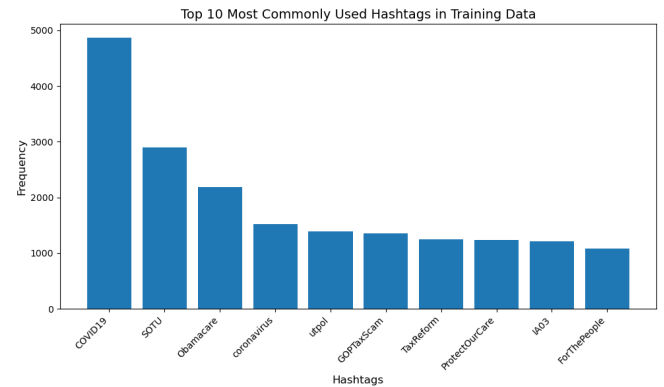


Fig. 2. Top 10 Most Common Hashtags Found in the Training Data

C. Summary Statistics - Training Data

Using the lemmatizer code, the tweet information in the training data was cleaned to find summary statistics for tweet length (characters), tweet length (words), hashtag length (characters), and hashtag length (words), as shown in Fig. 1.

We see that the variation in the character length of tweets and hashtags is greater than that of the word counts, respectively.

D. Most Common Hashtags - Top 10

Next, the training data was configured to identify the top ten most common hashtags, as shown in Fig. 2.

The top hashtags are related to the COVID-19 epidemic as well as political practices, which makes sense concerning both the time-period of the tweets (from 2008 to 2020) and the importance of the topics in United States society.

In the top 4 hashtags, we see 2 referring to the Coronavirus, emphasizing its significant impact on its severity and discourse. The other two, SOTU (State of the Union) and Obamacare relate to political events and policies, which seem to be key topics of social media discourse.

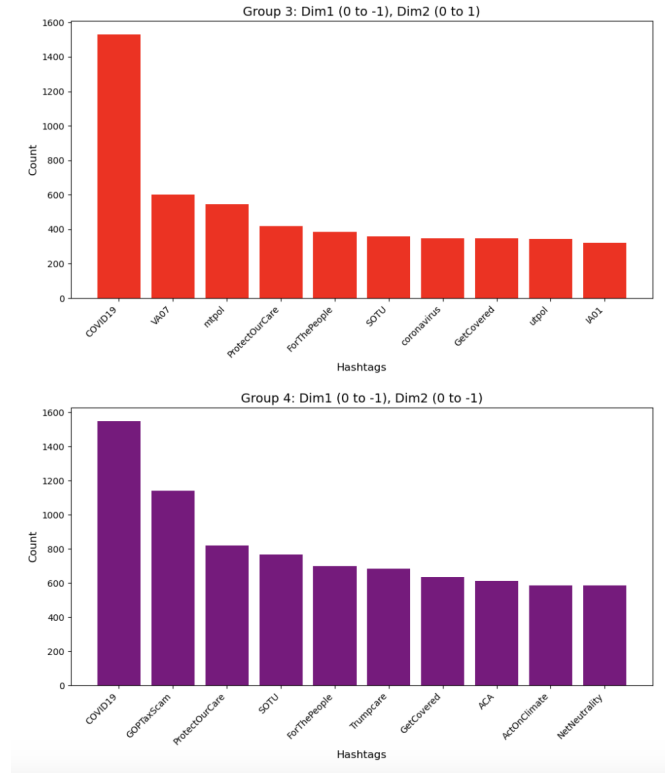
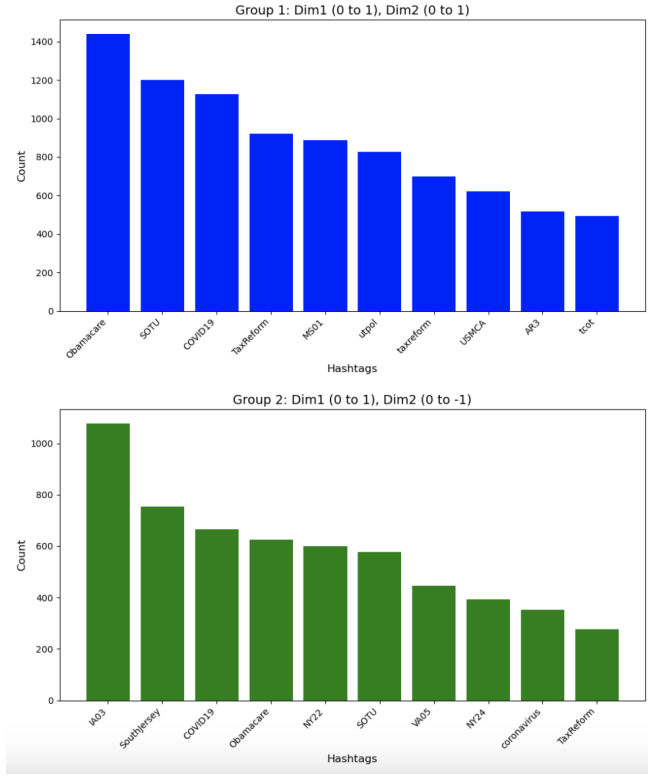


Fig. 3. Bar Charts for Different Groups of DW-Nominate Score's Most Commonly Used Hashtags

E. Division of Dataset into DW-Nominate Scores

Experimenting with the training dataset, the data was divided into four subgroups based on first and second ideology scores, as seen by Fig. 3. The findings are as follows:

- Group 1: There is greater domination from the political hashtags such as ObamaCare and SOTU, and the hashtags trending seem to be more liberal-leaning.
- Group 2: Unique hashtags dominate, such as IA03 and SouthJersey, indicating a focus on regionality over political and COVID-related tweets.
- Group 3: Most Common - COVID-19 along with a regional VA07 tweet, but the usage of ProtectOurCare and ForThePeople indicate a more liberal-focused group.
- Group 4: The mixture of ProtectOurCare, ActOnClimate, and TrumpCare shows a mixture of political affiliation in this group, but it is dominated by COVID-19 and GOPTaxScam.

F. Variation of 1st Dimension DW-Nominate Scores

Fig. 4. illustrates the change in the range of liberal-leaning (DW-Nominate Scores less than 0) and conservative-leaning (greater than 0) tweets.

Analysis over the years is consistent with each ideological group (conservative remains positive, liberal remains negative, with no overlaps, therefore meaning the division of nominate scores is "correct"). Conservatives are more stable in terms of density distribution, whereas liberals shift closer to 0 over the years, making their tweets overall seem trending towards political neutrality.

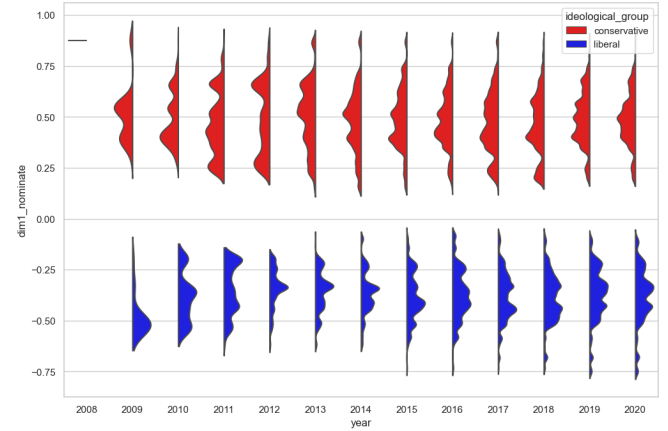


Fig. 4. Ridge Plot Showing the Variation of Liberal & Conservative Tweets over time (2008-2020)

G. Euclidean Distance of Ideologically Different Tweets

(NOTE: Due to time complexity, a sample of training data was extracted instead of the full dataset)

Using the Euclidean Distance formula on a sample training data set, tweet pairs were generated and the distance was calculated for both the first and second dimensions. From this, patterns were found, as seen in Fig. 5.

The top tweet pairs for both dimensions are the same as b. The top 10 most distant tweets are along the 1st dimension, therefore the first dimension contributes the most towards ideological differences. The lower distance values of the

Top 10 Most Distant Tweets (1st Dimension):

	Tweet1	Tweet2	Distance_Both	Distance_Dim1	Distance_Dim2
18738	39	58	1.876241	1.612877	0.958596
18786	39	106	1.876241	1.612877	0.958596
19090	39	410	1.876241	1.612877	0.958596
22382	47	58	1.876241	1.612877	0.958596
22430	47	106	1.876241	1.612877	0.958596
22734	47	410	1.876241	1.612877	0.958596
27396	58	166	1.876241	1.612877	0.958596
27447	58	217	1.876241	1.612877	0.958596
27612	58	382	1.876241	1.612877	0.958596
27660	58	430	1.876241	1.612877	0.958596

Top 10 Most Distant Tweets (2nd Dimension):

	Tweet1	Tweet2	Distance_Both	Distance_Dim1	Distance_Dim2
7993	16	146	1.712864	0.107448	1.709491
8329	16	482	1.712864	0.107448	1.709491
62382	146	260	1.712864	0.107448	1.709491
96291	260	482	1.712864	0.107448	1.709491
8060	16	213	1.653275	0.120848	1.648852
8210	16	363	1.653275	0.120848	1.648852
83755	213	260	1.653275	0.120848	1.648852
96172	260	363	1.653275	0.120848	1.648852
7940	16	93	1.780470	0.748159	1.615653
42295	93	260	1.780470	0.748159	1.615653

Top 10 Most Distant Tweets (Both Dimensions):

	Tweet1	Tweet2	Distance_Both	Distance_Dim1	Distance_Dim2
18738	39	58	1.876241	1.612877	0.958596
18786	39	106	1.876241	1.612877	0.958596
19090	39	410	1.876241	1.612877	0.958596
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Fig. 5. Top 10 Euclidean Distances for Tweets in 1st & 2nd Dimension Ideologies, as well as Both Dimensions

2nd dimension indicate that the divide is generally smaller, meaning that regional divides are less significant.

III. METHODS

A. Autogluon

To determine the model used for the Kaggle challenge, a sample of the training data was plugged into Autogluon's TabularPredictor class to find the model that fit the data with the lowest RMSE. The results found that LightGBM and WeightedEnsemble fit the 1st and 2nd dimensions best, respectively.

B. Data Pre-Processing

The given lemmatizer Python code as well as the tweet cleaner function created earlier was applied to both the training and testing data, particularly in the 'full text' column, to clean and dataset and fit it better for a model.

C. Defining Variables and Vectorization

The X training variable for the model is simply the new, cleaned text from the training data. The Y training variable would be the first and second DW-nominate scores.

Lastly, the text needed to be converted into numerical values to create a better fit for the model. Using TF-IDF vectorization techniques, the X training and X testing variables were converted into numerical values.

D. Model Decision/Hyperparameter Tuning

Due to its relative performance in Autogluon, the LightGBM model was chosen as the Regression technique for fit.

The best parameters were chosen through a combination of GridSearchCV, a hyperparameter tuning technique, as well as via the RMSE submission scores on Kaggle.

IV. RESULTS

A. Training and Prediction Timing

On average, the time it takes the model to both train and predict DW-Nominate Scores is approximately 25 minutes on a 2020 MacBook Pro, far beneath the required threshold.

B. Prediction Results

The best Kaggle score received is 0.25667, which is 8th place overall in the competition as of 11:13 AM on 02/09/2025.

C. Results Analysis

While the results are good and much better than the sample submission if given more time, there is room for further improvements to reduce RMSE.

- 1. Utilizing a Different Model: Instead of using a more traditional model, there are multiple models, such as DistilBERT, that are better techniques for datasets containing words. These techniques are likely better for the Twitter data given and could produce better results.
- 2. Further Hyperparameter Tuning: There is a time constraint with how many GridSearch runs/combinations can be completed, but given more time, further combinations could be trialed to produce better results.

- 3. Utilizing Different Preprocessing Techniques: Similar to the first improvement, instead of TF-IDF vectorization, other word/letter vectorizations could be used to generate better training data.
- 4. Increasing Memory Usage: Jupyter Notebook is limited in kernel memory, thus limiting the parameters used for this model. Using a PC and IDE with greater RAM and memory could help expand the preprocessing and model functionality.

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